Subdifferentials and Smoothed Gradient Descent

Reading notes and first experiments

November 2026

Background: gradient descent and SGD

- Goal: minimize $f(x)$, e.g. a training loss averaged over data, $f(x) = \frac{1}{N}\sum_{i=1}^{N} \ell_i(x)$.
- **Gradient descent:** $x_{k+1} = x_k - \gamma_k \nabla f(x_k)$.
- **SGD:** replace $\nabla f(x_k)$ with a cheap random estimate g_k (e.g. a minibatch gradient) that is right on average, $\mathbb{E}[g_k] = \nabla f(x_k)$:

$$x_{k+1} = x_k - \gamma_k\, g_k.$$

- Standard theory assumes f is **smooth**: ∇f is Lipschitz with constant L.
- Then SGD finds a point with $\mathbb{E}\|\nabla f(x)\| \le \varepsilon$ in about $1/\varepsilon^4$ steps. The rate depends on the smoothness constant and on the noise in g_k.

For nonconvex f, SGD guarantees a stationary point, not the global minimum.

Background: what we really do in practice

- In deep learning, f is the training loss of a network, and we run SGD or Adam with gradients from backpropagation.
- ReLU, max-pooling and ℓ_1 terms create **kinks**, so f is not smooth.
- At a kink, autograd just picks a value (e.g. $\mathrm{ReLU}'(0) = 0$), and we run SGD anyway.
- So the smooth theory doesn't technically apply. **What does SGD actually guarantee here?**
- Sometimes we don't have gradients at all: black-box models, or when backprop uses too much memory (e.g. zeroth-order fine-tuning of LLMs, MeZO). Then we only have **function values**.

This talk: a method with guarantees for nonsmooth f using only function values.

1. The problem

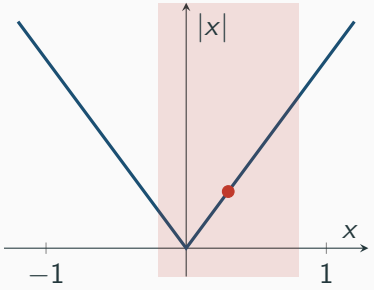
- Goal: minimize f that is Lipschitz but **nonsmooth** and **nonconvex** (e.g. training losses of ReLU networks).
- For nonconvex f we can't guarantee the global minimum, so we aim for a **stationary point**: no small step decreases f.
- Gradient descent stops when $\|\nabla f\|$ is small. That doesn't work here.
- For $f(x) = |x|$, the slope is ± 1 everywhere except exactly at 0, so no point is "almost stationary".
- Clarke subdifferential $\partial f(x)$ = convex hull of nearby limiting gradients. Getting ε-close to $0 \in \partial f(x)$ can't be guaranteed in finite time (Zhang et al. 2020).

We need a weaker notion of stationarity that we can actually reach.

2. Goldstein δ-subdifferential

$$\partial_\delta f(x) = \operatorname{conv}\Big(\bigcup_{\|y-x\|\le\delta} \partial f(y) \Big)$$

- Use subgradients from a whole δ-ball, not just at x.
- x is (δ,ε)-stationary if $\operatorname{dist}(0, \partial_\delta f(x)) \le \varepsilon$.
- This *can* be reached in finite time.
- As $\delta \to 0$ it becomes Clarke: $\bigcap_{\delta>0} \partial_\delta f(x) = \partial f(x)$.



$x = 0.3$, $\delta = 0.5$: the ball reaches the kink,
so $\partial_\delta f(x) = [-1, 1] \ni 0$.

3. Smoothing

$$f_\nu(x) = \mathbb{E}_\xi\big[f(x+\nu\xi)\big] \qquad (\xi \text{ uniform in the unit ball})$$

- Averaging f over a small ball removes the kinks.
- $|f_\nu - f| \le L\nu$, and ∇f_ν is Lipschitz with constant $\approx \sqrt{n}\,L/\nu$.
- Gradient estimate from only **two function values** (v uniform on the sphere):

$$g = \frac{n}{2\nu}\big(f(x+\nu v) - f(x-\nu v)\big)\,v, \qquad \mathbb{E}[g] = \nabla f_\nu(x).$$

- Trade-off: small ν is accurate but harder to optimize.

4. Key fact

$$\nabla f_\nu(x) \in \partial_\nu f(x)$$

- $\nabla f_\nu(x)$ is the **average of the gradients** at points within ν of x.
- An average of those subgradients lies in their convex hull: the Goldstein set.
- So $\|\nabla f_\nu(x)\| \leq \varepsilon \implies x$ is (ν, ε)-stationary.
- SGD on the smooth f_ν gives a guarantee for the nonsmooth f, using only function values: roughly $n^{3/2}/(\delta\varepsilon^4)$ evaluations (from the paper).

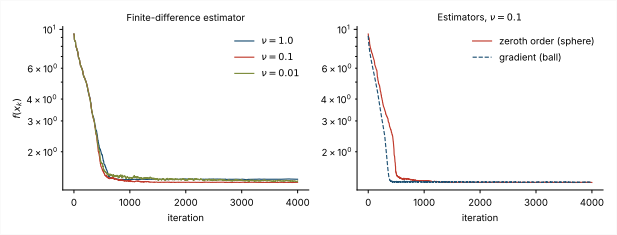
5. What I tried

Python implementation on

$$f(x) = \frac{1}{m}\|Ax-b\|_1+\lambda\sum_j \min\{|x_j|,\tau\}$$

(nonsmooth, nonconvex), $n = 20$, 4000 steps.

ν	$f(x_T)$	$\|\nabla f_\nu(x_T)\|$
1.0	1.39	0.02
0.1	1.34	0.07
0.01	1.36	0.95



Same objective for every ν, but with tiny ν the smoothed gradient stays large.

6. Question I'd like to work on

- The guarantee is for a **fixed** ν.
- Shrinking $\nu \to 0$ gives Clarke stationarity in the limit, but no rate.
- Idea: shrink it during the run, $\nu_k = \nu_0/(k+1)^\alpha$, with matching step sizes.

Can we get a rate toward Clarke stationarity using only function values?

Next steps

1. Try a few values of α on the same problem and plot $\|\nabla f_{\nu_k}(x_k)\|$.
2. Check what's already known in the literature.

Summary

1. For nonsmooth, nonconvex f, Clarke stationarity can't be reached in finite time.
2. The Goldstein δ-subdifferential is a reachable relaxation, and it becomes Clarke as $\delta \to 0$.
3. Smoothing f over a ball of radius ν gives a smooth function whose gradient can be estimated from two function values.
4. Key fact: $\nabla f_\nu(x) \in \partial_\nu f(x)$, so a small smoothed gradient means (ν, ε)-Goldstein stationarity.
5. In my experiment, every ν reached a similar objective, but small ν left the smoothed gradient large.
6. Next: shrink ν during the run and look for a rate toward Clarke stationarity.